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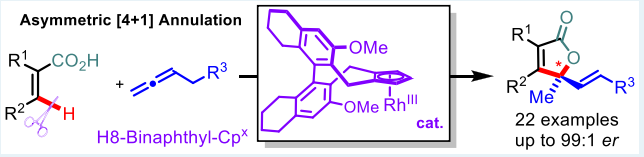
Asymmetric Cp^{x}Rh(III)-Catalyzed Acrylic Acid C−H Functionalization with Allenes Provides Chiral γ-Lactones

Shou-Guo Wang and Nicolai Cramer*

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ABSTRACT: A chiral cyclopentadienyl (Cp^x) ligand featuring the semi-saturated H8-binaphthyl backbone as the chiral element was developed. Its application potential was demonstrated by Cp^{x}Rh^{III}-catalyzed C−H bond functionalization of acryl acids and allenes, exhibiting superior catalytic performance compared to other Cp^x ligands. The outlined enantioselective [4 + 1]-annulation process results in synthetically attractive α,β-unsaturated γ-lactones having an allylic stereogenic center, which were obtained in enantioselectivities of up to 99:1 enantiomeric ratio.



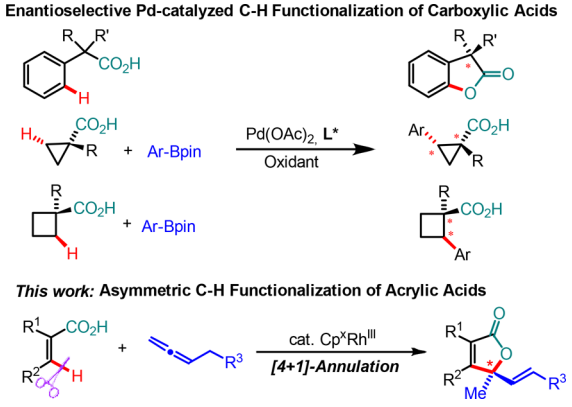
KEYWORDS: *asymmetric catalysis, rhodium, C−H activation, chiral Cp ligands, [4 + 1] annulation*

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Carboxylic acids are very widespread, easily available, and relatively inexpensive making them attractive and widely used intermediates in organic synthesis. In the past, direct C−H functionalizations of free carboxylic acids have attracted significant attention, because of the powerful and straightforward access to diverse important fundamental building blocks.[1] Compared to amide- or hydroxamate-directed enantioselective C−H functionalizations,[2] corresponding reactions of free carboxylic acids are relatively underdeveloped. In 2013, Wang and Yu reported an asymmetric Pd(II)-catalyzed lactonization of arylacetic acid to access chiral benzofuranones via acid-directed aromatic C(sp^2)−H activation.[3] Later, Yu and co-workers developed Pd-catalyzed acid-directed enantioselective C(sp^3)−H activation/cross-coupling reactions of cyclopropane- or cyclobutane−carboxylic acids.[4] Despite this progress, many challenges for efficient enantioselective C−H functionalization of free carboxylic acids remain unsolved. In this respect, the use of acrylic acids is very attractive from the point of view of availability and utility and is challenging, because the acids are frequently prone to polymerizations,[5] and they act as acceptor olefins in such C−H functionalizations.[6] Sparked by our long-standing interest in asymmetric functionalizations of challenging alkenyl C−H bonds[7] and inspired by Cheng's acrylic acid directed Rh^{III} functionalizations,[8] we aimed to leverage chiral Cp^{x}Rh^{III} complexes as catalysts for enantioselective alkenyl C−H functionalization of acrylic acids with allenes (Scheme 1). Such [4 + 1] annulation process would provide a very efficient enantioselective access to functionalized α,β-unsaturated-γ-lactones with quaternary stereogenic center. This structural motif is prevalent in a range of natural products and active compounds with diverse biological activities (see Figure 1).[9]

The development of powerful chiral Cp ligands (Cp^x) plays a vital role for Cp^{x}Rh^{III} axis of enantioselective C−H functionalizations.[10] Since Ward and Rovis,[11] and our

Scheme 1. Carboxylic-Acid-Directed Asymmetric C−H Functionalizations



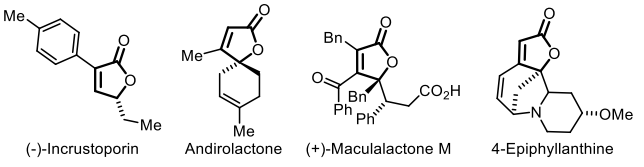
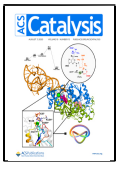


Figure 1. Selected natural products with an α,β-unsaturated-γ-lactone substructure.

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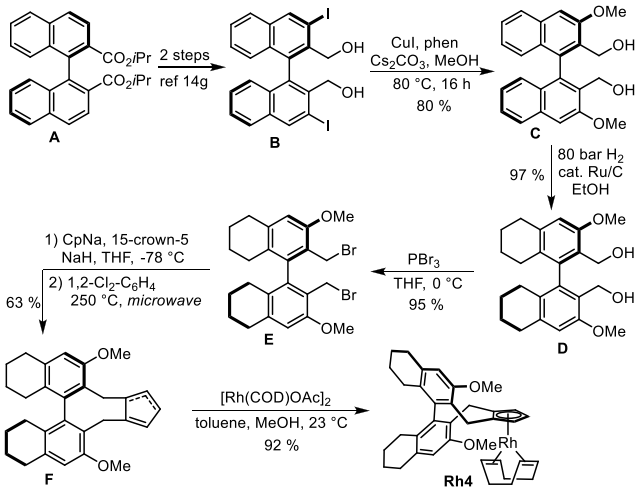
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group[12] independently reported applications of chiral Cp^xRh^{III} catalysts for asymmetric C−H functionalizations, a rapid development was triggered.[13] Over the past years, our group,[14] as well as You and co-workers,[15] Antonchick and Waldmann, and co-workers,[16] and Perekalin and co-workers[17] disclosed different Cp^x ligand families and successfully applied in asymmetric catalysis. Despite the advances on chiral Cp ligands, the efficient access to new chiral Cp ligands with different backbones and selectivity properties remains in very high demand. Compared to the binaphthyl backbone catalysts its chiral semi-saturated H8-binaphthyl analogue possesses an alternative conformational rigidity, as well as different geometry and dihedral angle, influencing its performance in asymmetric inductions.[18] Therefore, we aimed for a novel Cp^x ligand bearing the rigid axial chiral H8-binaphthyl backbone to boost enantioinductions, compared to the parent binaphthyl-based Cp^x versions.

Leveraging our previously developed efficient route to diiodide **B**,[14g] subsequent Ullmann coupling provided a 3,3-dimethoxy core **C** in 84% yield on a gram-scale (see Scheme 2). Exploration of several hydrogenation catalysts and

Scheme 2. Synthesis of H8-Binaphthyl Cp^x Ligand and Its Corresponding Cp^xRh^I Complex[a]



[a]phen = 1,10-phenanthroline.

conditions, targeted semihydrogenated H8-binaphthyl intermediate **D** could be obtained in 97% yield from **C**, employing Ru/C as a hydrogenation catalyst under a H_2 pressure of 80 bar in EtOH. Treatment of **D** with PBr_3 smoothly converted both benzylic hydroxy groups to the corresponding dibromide **E** in 95% yield. Double alkylation of CpNa with **E** and subsequent exposure of the formed products to thermal rearrangement gave designed chiral H8-binaphthyl Cp^x ligand **F**. The corresponding chiral $FRh^I(COD)$ complex (**Rh4**) was smoothly obtained by our complexation strategy, using $[Rh(COD)OAc]_2$ in toluene/methanol at ambient temperature under air.[19]

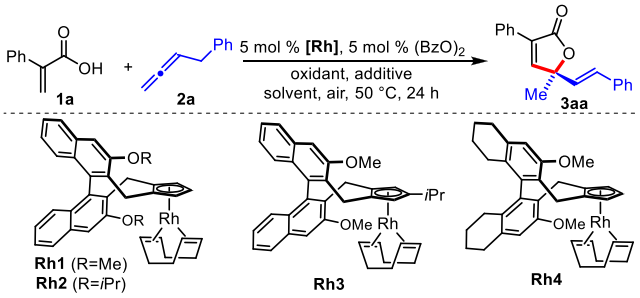
The enantioselective alkenyl C−H transformation [4 + 1] annulation process was studied and initially optimized with 2-phenylacrylic acid (**1a**) and benzylallene (**2a**) as a substrate combination (see Table 1). Using complex **Rh1** having the standard MeO-binaphthyl-derived Cp^x ligand oxidatively activated by benzoyl peroxide as a catalyst in *t*BuCN as a solvent and $Cu(OAc)_2$ as a terminal oxidant gave rise to targeted lactone **3aa** in 38% yield and 92:8 er (Table 1, entry 1). Increased bulk in the ligand backbone using −O*i*Pr-substituted Cp^x ligand (**Rh2**) delivered essentially similar result (Table 1, entry 2). **Rh3** having a trisubstituted Cp^x ligand significantly improved the asymmetric induction and gave **3aa** in 95:5 er, although in much lower yield (19%) (see Table 1, entry 3). Notably, complex **Rh4** equipped with the newly developed chiral H8-binaphthyl Cp^x ligand **F** displayed a highly favorable catalytic performance in this asymmetric vinyl C−H transformation. **Rh4** led to a remarkable improvement on both catalytic efficacy and enantioselectivity, forming **3aa** in 49% yield and 95:5 er (Table 1, entry 4). A brief screening of different nitriles as solvents showed that *t*BuCN provides the best overall performance in yield and selectivity (see Table 1, entries 4−6). Acetonitrile gave a higher yield (58%) but an enantiomeric ratio (er) of only 87:13 (Table 1, entry 5), evidencing the strong and intimate effect of the nitrile solvent in the reaction. Different copper oxidants had little influence on the reactivity and the enantioselectivity of the process (Table 1, entry 7). Increasing the amount of $Cu(OAc)_2$ to 2 equiv reduced the yield of **3aa** to 40% (see Table 1, entry 8). Considering the modest reaction performance might be attributed to losses of acrylic acid **1a** and, to some extent of allene **2a** by competitive reaction pathways, including conjugate additions, radical reactions and mainly polymerization,[20] different additives (TEMPO, BHA and BHT) were then examined to mitigate these pathways (Table 1, entries 9−11). Interestingly, employing BHT as the additive, γ-lactone **3aa** formed with improved yield while still maintaining the excellent enantioselectivity (Table 1, entry 11). An increased catalyst loading and higher concentration enabled the enantioselective synthesis of functionalized γ-lactone product **3aa** in good yield and 95:5 er (Table 1, entry 12).

With the optimized conditions, we examined the generality of the [4 + 1] annulation transformation, with respect to acryl acids **1***x* (Scheme 3). The influence of steric and electronic effects of the aryl group on the α-position of the acryl acids (R^1) was evaluated on the reaction performance with benzyl allene **2a**. Aromatic rings bearing electron-donating or electron-withdrawing substituents were competent in this transformation, generating γ-lactone products **3aa**−**3ha** in moderate to good yields and with excellent enantiocontrol (yield = 40%−72% and er = 94:6−95.5:4.5). Sterically demanding aryl substituents such as 1-naphthyl and *o*-tolyl displayed a good performance in the [4 + 1] annulation reaction, delivering corresponding products **3ja** (71% yield, 94:6 er) and **3ka** (62% yield, 92.5:7.5 er) with good yields and selectivity. Benzyl-substituted substrate **1l** was suitable and resulted in lactone **3la**. In addition, tricyclic lactone **3ma** was generated from cyclic 2,3-disubstituted acryl acid **1m** in good yield and albeit slightly decreased er. However, cinnamic acid-type substrates were found to be much less reactive, indicating that a substituted group at α-position of the acrylic acid substrate should play an important role to facilitate the carboxylate-directed CMD-type metalation of vinyl C−H bond with the chiral Cp^xRh catalyst.

The transformation shows compatibility as well with a wide range different allenes **2***y* (see Scheme 4). A series of substituted benzyl allenes with a variety of aryl groups with different electronic and steric properties are competent in the vinyl C−H activation process providing access to lactones **3ab**−**3af** in good yields and excellent enatioselectivity (yield = 60%−75%, er = 94:6−99:1). Notably, *o*-Me benzyl allene **2b**

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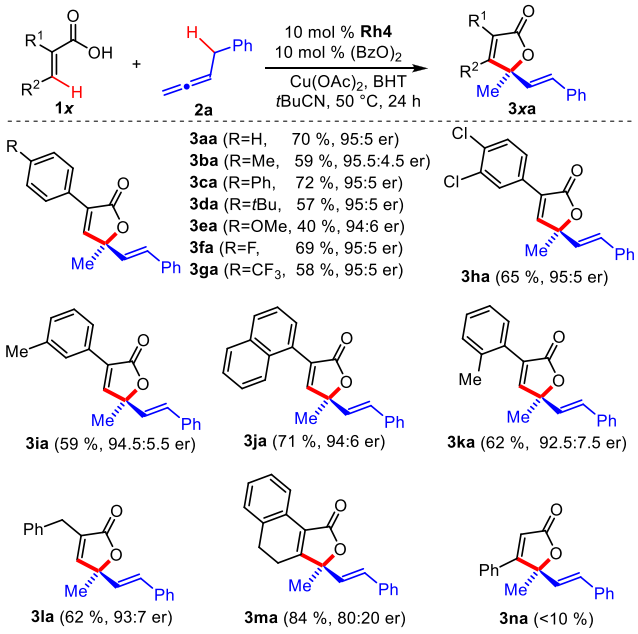
Table 1. Optimization of the Enantioselective Acrylic Acid C−H Functionalization[a]



entry	[Rh]	solvent	oxidant	additive	yield[b] (%)	enantiomeric ratio, er[c]
1	**Rh1**	*t*BuCN	$Cu(OAc)_2$	–	38	92:8
2	**Rh2**	*t*BuCN	$Cu(OAc)_2$	–	40	91:9
3	**Rh3**	*t*BuCN	$Cu(OAc)_2$	–	19	95:5
4	**Rh4**	*t*BuCN	$Cu(OAc)_2$	–	49	95:5
5	**Rh4**	MeCN	$Cu(OAc)_2$	–	58	87:13
6	**Rh4**	*i*PrCN	$Cu(OAc)_2$	–	38	91:9
7	**Rh4**	*t*BuCN	$Cu(OBz)_2$	–	48	94:6
8[d]	**Rh4**	*t*BuCN	$Cu(OAc)_2$	–	40	95:5
9[e]	**Rh4**	*t*BuCN	$Cu(OAc)_2$	TEMPO	46	95:5
10	**Rh4**	*t*BuCN	$Cu(OAc)_2$	BHT	55	95:5
11	**Rh4**	*t*BuCN	$Cu(OAc)_2$	BHA	52	94:6
12[f]	**Rh4**	*t*BuCN	$Cu(OAc)_2$	BHT	70	95:5

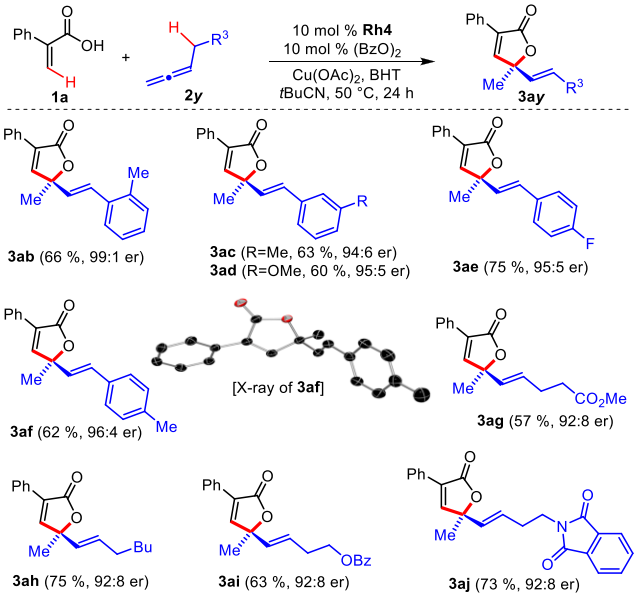
[a]Conditions: 50 μmol **1a**, 100 μmol **2a**, 5 mol % [**Rh**], 5 mol % additive, 25 μmol oxidant, 0.1 M in solvent at 50 °C for 24 h. [b]Isolated yield of **3aa**. [c]As determined by chiral HPLC. [d]With 100 μmol oxidant. [e]10 mol % TEMPO. [f]With 10 mol % [**Rh**], 10 mol % additive, 0.5 equiv oxidant, 2 equiv BHT, 0.2 M. TEMPO = (2,2,6,6-tetramethylpiperidin-1-yl)oxyl, BHA = 2-*tert*-butyl-4-methoxyphenol, BHT = 2,6-di-*tert*-butyl-4-methylphenol.

Scheme 3. Scope of the Acrylic Acids 1 for the Formation of Asymmetric Lactone[a]



[a]Conditions: 0.1 mmol **1***x*, 0.2 mmol **2a**, 10 μmol **Rh4**, 10 μmol $(BzO)_2$, 0.05 mmol $Cu(OAc)_2$ in 0.5 mL *t*BuCN at 50 °C for 24 h.

Scheme 4. Scope of Allenes 2 for the Asymmetric Lactone Formation[a]



[a]Conditions: 0.1 mmol **1a**, 0.2 mmol **2y**, 10 μmol **Rh4**, 10 μmol $(BzO)_2$, 0.05 mmol $Cu(OAc)_2$ in 0.5 mL *t*BuCN at 50 °C for 24 h.

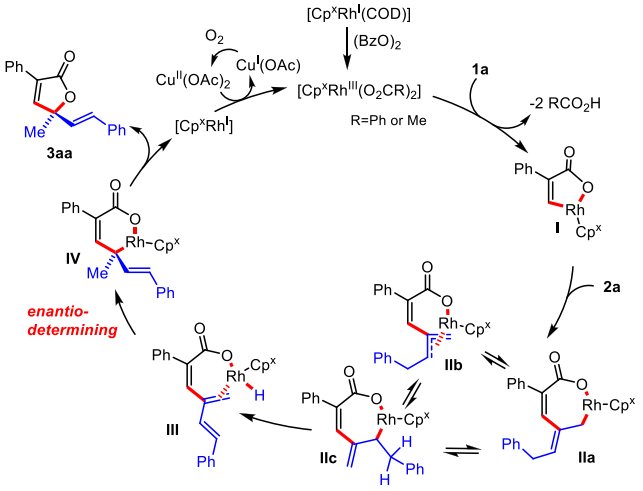
resulted in lactone **3ab** with an excellent enantiomeric ratio of 99:1, possibly influenced by the steric hindrance of this substrate. Further nonaryl bearing aliphatic allenes were suitable for the [4 + 1] annulation, indicating that a conjugated olefin in the product is not a required driver for the overall transformation. Various functional groups, such as esters and phthalimido groups, were well-tolerated and afforded target lactones **3ag**−**3aj** in good yields and high enantioselectivity. X-ray crystallographic analysis of lactone **3af** allowed the absolute configuration to be determined to be (*S*).[21] Configurations of the products **3** were assigned by analogy.

The following catalytic cycle for the asymmetric alkenyl C−H functionalization of acryl acids with allenes is proposed (see

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Scheme 5).[7b,8,15e] The Cp^xRh^I precatalyst enters the catalytic cycle via oxidation with $(BzO)_2$ forming the active Cp^xRh^{III}

Scheme 5. Proposed Catalytic Cycle for the Rh^{III}-Catalyzed Enantioselective [4 + 1] Annulation of Acrylic Acids and Allenes



species. Acid exchange with substrate **1a** and subsequent CMD-type directed alkenyl C−H activation forms five-membered rhodacycle **I**. Insertion of allene **2a** results in allyl rhodium intermediates **IIa**−**IIc**. β-Hydride elimination generates triene intermediate **III**. Subsequent enantio-determining hydrorhodation results in allyl rhodium species **IV**. Reductive elimination delivers lactone **3aa** and generates a Cp^xRh^I species. The catalytic cycle is closed by its oxidation of $Cu^{II}(OAc)_2$ to form the Cp^xRh^{III} complex. Since the process is conducted under an air atmosphere, Cu^{II} is regenerated by oxygen.

In summary, we have developed a chiral Cp^x ligand using the semi-saturated H8-binaphthyl backbone as the chiral element. The potential of this new ligand was showcased with the corresponding Cp^xRh complex exhibiting superior catalytic performance, compared to existing binaphthyl-derived Cp^x ligands, in challenging C−H bond functionalization of acryl acids. The rhodacyclic intermediate engages in an enantioselective [4 + 1] annulation process with a wide range of allenes. Highly functionalized and synthetically attractive unsaturated γ-lactones having an allylic stereogenic center were obtained in enantioselectivities of up to 99:1 er.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c02430.

Synthetic procedures, characterization data for all new compounds (PDF)

Crystallographic data for **3af** (CIF)

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Notes

The authors declare no competing financial interest.

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